Andrea Gallegati, Ph.D

CSSLP – DevSecOps Engineer @ IDS AirNav – ENAV Group

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EXPERIENCE

IDS AirNav – ENAV Group

DevOps Engineer

Jul 2024 – Ongoing | Rome, ITA

- Managing several (backend) microservices, shared by any product in the suite, and some web apps, essential for the functionality and management of it.
- Maintaining and frequently updating third-party components to address rapidly emerging vulnerabilities.
- Developing IaC and maintaining an installation and configuration tool based on Ansible, to standardize the deployment of solutions.
- Enabling a stable foundation that accelerates development by offering standardized tools, services, and integration patterns across the organization.
- Supporting operations and assisting customers when issues escalate beyond the third level of support.

JHipster | Ansible | Postgres | Keycloak | DevSecOps | Docker Swarm | Postman | Jenkins | Burp Suite

Security Champion

Mar 2023 – Ongoing | Rome, ITA

- Certified Secure Software Lifecycle Professional (CSSLP – ISC2). Undergoing continuous training in cybersecurity.
- Assisting operations team in the deployment of containerized solutions in clustered environments, with Ansible.
- Leading threat modeling, dependency analysis, and security assessments (SAST/DAST findings, ASVS requirements), using advanced Jenkins pipelines to ensure a seamless CI/CD process.

CSSLP | Threat Modeling | CIS | ASVS | OWASP | SCA | SAST | Defect Management | CI/CD

Software Engineer

Apr 2021 – Jul 2024 | Rome, ITA

- Developing an high-load web application (for aeronautical publishing) using Spring Boot. A product sold by ENAV Group, the Italian Air Navigation Service Provider (ANSP) with hundreds of customers all around the world.
- Troubleshooting some performance issues, sometimes refactoring its microservice architecture, built around Apache Kafka, consuming massive data.
- Maintaining the legacy IAM solution (developed in-house) and taking part to the transition toward the current IAM solution (based on Keycloak).
- Designing and developing a migration tool for the transition from the legacy IAM solution. I made it in python, using Django for the migration tool dashboard.
- Organized two editions of a small hackathon at Sapienza University, using my network in the academy and challenging students with programming tasks. This brought new talent into our company.

Java | Spring Boot | Apache Kafka | Django | Non-Functional Requirements | Microservices | Vaadin

Temple University – Rome Campus

Intern Supervisor

May 2024 – Ongoing | Rome, ITA

- Sponsored student internships at Edukai, a startup where they developed and integrated services into the core product using advanced generative AI techniques and open-source projects in innovative ways.
- Supervised an intern working on the "Tiber River" project, developing a pipeline (in *Python*) to automate the generation of 3D models of iconic monuments using OpenStreetMap data (OSM), advanced photogrammetry with generative AI, and automated 3D modeling refinements.

Adjunct Instructor

Jan 2019 – Ongoing | Rome, ITA

- Teaching several classes, including *Problem Solving & Programming in Python*, *Intermediate Algebra*, a foundational course that is part of the GenEd (General Education) curriculum and *Mechanics of Solids*.
- Organized annually the *Motor Valley* field trip to Modena, with workshops, seminars and meetings with motorsport experts (i.e. Dallara Academy, the Maserati factory and the Ferrari museum) for the *Solids* class.

Educational Program Design | Lesson Planning | University Teaching | Online Tutoring

Teaching Assistant

Jan 2018 – Dec 2019 | Rome, ITA

- Teaching *Mechanics of Solids* in English, to American mechanical engineering students.
- Gained significant insight into the Anglo-Saxon approach to education.

Cybertech.eu – Engineering

System Analyst Intern

Apr 2021 – Apr 2021 | Rome, ITA

- Analyzing systems to implement appropriate Identity Access Management (IAM) solutions for a variety of clients in both Linux and Windows environments.

Accenture

Software Developer Intern

Mar 2021 – Mar 2021 | Rome, ITA

- Developing a web-based management software for a department of the Italian Ministry of the Interior.
- Working on jBPM (Java Business Process Model) flows, to automate tasks and managing process instances.

Java Academy

Dec 2020 – Feb 2021 | Rome, ITA

- Crash Course in Java (by Accenture), providing a significant leap into the world of software dev for technology.

Sapienza University

PhD Researcher

Sep 2017 – Jan 2021 | Rome, ITA

- Conducting my research project, mentoring a few students each semester, that contributed to my own project.

Research | Team Leading | Mentoring | Git | Agile

Teaching Assistant

Mar 2019 – Jun 2020 | Rome, ITA

- Teaching assisting a computational aerodynamics lab, for the Department of Aerospace and Mechanical Engineering (DIMA), in a small class environment focused on hands-on learning.
- Training students on programming and applying basic aerodynamic principles to three-dimensional finite wings design. The code was developed in groups.
- Scouting students to recruit them, getting involved in departmental projects and join our research team.

Teaching Assistant

Oct 2018 – Dec 2019 | Rome, ITA

- Teaching assisting a calculus class for the School of Industrial Engineering (ICI) to hundreds of freshman aerospace engineering students. Some of my former students are now working with me.
- Teaching assisting a Linear Algebra class for the Department of Basic Sciences Applied to Engineering (SBAI) to Biomedical, Energy, and Electrical Engineering classes.
- Interacting with students from different academic backgrounds and coordinating with different professors, with different teaching methods.

PROJECTS

Temple University - Rome Campus

walkintiber - Immersive Digital Twin of Rome

Aug 2024 – Present Rome, ITA

- Co-founded a high-resolution, 3D digital twin of Rome accessible on the web, allowing users to virtually explore the Eternal City in immersive detail.
- Automated 24/7 pipeline to integrate live community data from OpenStreetMap (OSM), scaling the infrastructure horizontally for global coverage and vertically for enriched realism, including: textures, digital elevation models (DEM), and environmental detail.
- Integrating generative AI for texture creation and prototyped monument modeling using photogrammetry; developing VR compatibility for enhanced user engagement.
- Engaging TUR students and faculty through local "walks in Rome" contributing to OSM, and promoting hands-on learning in data mapping, AI, and VR technologies.

This project demonstrates TUR's commitment to innovative educational experiences that blend advanced technology with cultural heritage.

IaC | Terraform | MongoDB Atlas | FastAPI | NestJS | Cloudflare | Apache Airflow | Blender | PlayCanvas

IDS AirNav – ENAV Group

Migration Tool - Seamless transition from a legacy IAM solution

Apr 2023 – Oct 2023 Rome, ITA

- Designed the migration tool, consisting of two microservices: a Django web application for the dashboard, and Redis for real-time logging capabilities, leveraging Channels (within Django) and eliminating the need for polling.
- Automated the migration process to facilitate a seamless transition from a legacy IAM solution to a modern Keycloak-based IAM system, ensuring minimal downtime and business continuity for users.
- Provided two deployment options: a full-featured version with a graphical interface for configuration and tracking, and a lightweight "headless" version allowing CLI-driven migrations.
- Integrated a bunch of different clients (HTTP, LDAP, PostgreSQL, and SSH using *paramiko*) for accessing and retrieving configurations from the legacy system, ensuring validation features on the dashboard itself.

This not only provided a structured approach to the transition, but also enabled detailed tracking and verification, ensuring data integrity and system resilience.

NASA SpaceApps Challenge

CROPP - A whole new way of farming

Apr 2015 - Oct 2015 Rome, ITA

- First-prize winning project. Easy-to-use platform, for farmers, to monitor and address crop-related threats.
- On-field sensors embedded in distributed hardware devices. Imagery from Low Earth Orbit (LEO) satellites.
- A 48-hour hackathon, to develop a working prototype for the IoT device and an Android mobile app.
- Demo at the World Expo hosted by Milan, in Italy, themed "Feeding the Planet, Energy for Life."
- Pitched the project to NASA's then-Administrator, Charles Bolden Jr. at the U.S. Pavilion.

TALKS

IDS AirNav – ENAV Group

AeroSIG 20th Edition

11 – 12 Oct 2022 Rome, ITA

The Aeronautical Special Interest Group (AeroSIG) conference gathered colleagues and industry experts from over 35 countries, featuring a plenary session on the evolution of IDS AirNav solutions and breakout sessions focused on advanced topics in Air Traffic Management (ATM) and Aeronautical Information Management (AIM).

Presentation title: "IAM Status and Updates."

- Outlining the transition from the legacy IAM solution to a new system, based on Keycloak and customizations.
- Addressing IAM modernization strategies, detailing how to leverage Keycloak for improved security/interoperability.

AIMETA - Ass. Italiana Meccanica Teorica e Applicata

XXIV Congress AIMETA 2019

15 - 19 Sep 2019 | Rome, ITA

The AIMETA Congress is a prominent event fostering cross-disciplinary collaboration in mechanics, mathematics, and applied sciences. The event is recognized for bridging Rational Mechanics and Mathematical Physics with applications in fields such as biomechanics and material science.

Presentation title: "Droplet Homogeneous Nucleation in Two-Way Coupled Turbulent Flows"

EUROMECH Conference

ETC17 - European Turbulence Conference

3 - 6 Sep 2019 | Turin, ITA

The European Turbulence Conference (ETC) is a bi-annual event providing a platform for over 450 scientists to present recent advances in turbulence research. ETC17 gathered experts in flow dynamics, offering insights into cutting-edge developments in the field.

Presentation title: "Droplet Nucleation in Turbulent Steam Jets"

ERCOFTAC Workshop

DLES12 - Direct and Large Eddy Simulation Workshop

5 - 7 Jun 2019 | Madrid, ESP

The ERCOFTAC DLES Workshop is dedicated to advancements in turbulent flow simulation, focusing on Direct Numerical Simulation (DNS) and Large Eddy Simulation (LES) techniques. The 2019 edition gathered specialists to discuss advances in computational modeling of transitional and turbulent flows.

Presentation title: "Droplet Homogeneous Nucleation in a Turbulent Steam Jet in the Two-Way Coupling Regime"

PUBLICATIONS

Conference Proceedings

- **A. Gallegati**, F. Battista, P. Gualtieri, and C. M. Casciola, "Droplet homogeneous nucleation in a turbulent vapour jet in the two-way coupling regime," in *Direct and Large Eddy Simulation XII*, M. García-Villalba, H. Kuerten, and M. V. Salvetti, Eds., Cham: Springer International Publishing, 2020, pp. 143–149, ISBN: 978-3-030-42822-8.

AWARDS

Fondazione Roma Sapienza

Excellent graduate student - 6th edition

16 Apr 2018 | Rome, ITA

Awarded among the best graduates of 2017 at Sapienza.

NASA - Kennedy Space Center (KSC)

Galactic Impact Prize

2 Sep 2015 | Cape Canaveral, USA

Awarded and invited to attend a scheduled rocket launch.

GRANTS

Temple University

Professional Development Grant

Oct 2024 - Jul 2025 | Rome, ITA

Funding to support research and creative work aimed at promoting interdisciplinary learning and global engagement. This involves the application of advanced technologies such as generative AI and VR.

CINECA

ISCRA Class C Projects - HP10CEKU6J

May 2020 - Feb 2021 | Rome, ITA

ISCRA Class C Projects - HP10CQBSIR

Oct 2018 - Nov 2019 | Rome, ITA

Resources on HPC facilities for advanced numerical simulations in fluid dynamics, leveraging up to 100,000 core hours for heavy workloads and visualization tasks.

EDUCATION

Sapienza University – DIMA

Ph.D. in Theoretical & Applied Mechanics

Nov 2017 – May 2021 | Rome, ITA

Thesis title: "Droplet condensation in turbulent jets." Supervised by: Prof. Paolo Gualtieri.

- Developing high-performance software (HPC) for simulating complex, high-resolution, multi-phase turbulent flows.
- Maintaining optimized codebases, using parallel computing paradigms (OpenMP, MPI, CUDA) to take full advantage of distributed computing facilities
- Deriving an analytical approach to accurately represent phase transition processes interaction, in turbulent flows.
- Building a pipeline to deploy the compiled code onto remote supercomputers, running simulations on different environments and to periodically post-process intermediate results, monitoring the status real-time.

Turbulence | Multiphase Flows | HPC | Direct Numerical Simulations | Fortran

M.Sc. in Aeronautical Engineering

Sept 2014 – July 2017 | Rome, ITA

110/110 cum laude Thesis title: "A consistent framework for mass, momentum and energy exchange in two phase flows: DNS of a turbulent jet in the two-way coupling regime." Supervised by: Prof Carlo Massimo Casciola.

- Specialized into aerodynamics, propulsion, and structures (APS) curriculum.
- Focused on continuum mechanics, applying fundamental science to solve complex engineering problems.
- Fascinated by theoretical approach and numerical methods.

Fortran | PDE | Navier-Stokes Eqs.

ISAE Supaero – DAEP

Erasmus+ Program

Jan 2016 – Aug 2016 | Toulouse, FRA

Project title: "Cost versus accuracy in LES of wall bounded flows in CharlesX." Supervised by: Prof. Julien Bodart.

- Studied in an international environment, in the heart of an aerospace hub in Europe, with many different cultures.
- Experienced with enthusiasms the multidisciplinary approach, that is typical of France and that has enriched me so much from a technical point of view.
- Developed Python code for post-processing fluid dynamics numerical simulations, benchmarking a software developed by the Center for Turbulence Research (CTR), in Stanford.

python | C++ | bash | French

Sapienza University

B.Sc. in Aerospace Engineering

Sept 2011 – Nov 2014 | Rome, ITA

110/110 cum laude Thesis title: "The Newton's problem of minimal resistance." Supervised by: Prof. Andrea Dall'Aglio.

Functional Analysis | LaTeX

LSS J.F. Kennedy

High School Diploma

Sept 2006 – July 2011 | Rome, ITA

87/100 Baccalaureate thesis: "Revolutions in history and scientific thought."

- Playing competitive water polo with daily training. I learned a lot about cooperation, discipline and resilience, both in and out of the pool.
- Advanced education program (PNI), including intensive study of scientific subjects such as mathematics and physics, complemented by the use of computer science as a tool.

REFEREES

Prof. Andrea Dall'Aglio @ Sapienza University
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Prof. Mary Conran @ Temple University Rome Campus
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Prof. Julien Bodart @ ISAE Supaero
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LANGUAGES

English (Full Professional)
French
Italian (Native)

MORE

I love working in diverse teams, fostering growth, and building strong relationships with my colleagues.

Outside of work, you'll find me cooking with friends and family, discussing movies of all genres, staying active, or diving into late night coding sessions.

I'm always looking for opportunities to grow, share knowledge, and make a positive impact. I free-dive into the blue and look around. I like to take action.

Last updated: November 18, 2024